For Examiner's Use

2 A golf ball is hit from point A on the ground and moves through air to point B. The path of the ball is shown in Figure 2.1.

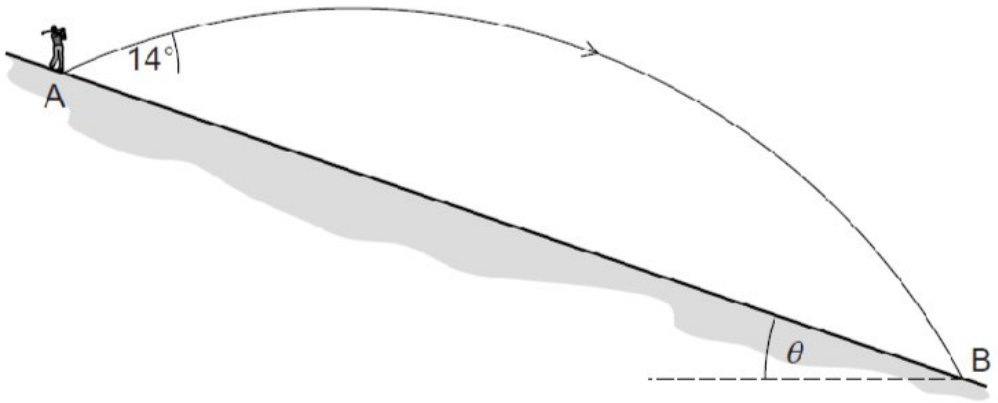


Figure 2.1

The ground slopes downhill and makes an angle θ to the horizontal as shown. The ball has an initial velocity of 63 m s^{-1} at an angle of 14° to the horizontal. It hits the ground at B after 4.9 s.

(a) Assuming air resistance to be negligible, calculate

(i) the vertical displacement from A to B,

vertical displacement = ... m [2]

(ii) the angle θ of the slope to the horizontal.

θ = ... ° [2]

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(b) In a real situation, air resistance is not negligible.

(i) On Figure 2.1, sketch the path of the golf ball when air resistance is not negligible. [2]

(ii) Explain how the maximum vertical height of the golf ball is affected.

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..... [1]

[Total: 7]